

VTS Motor Vehicle Challenge 2027: Management of Energy Efficiency and Safe Operation for Multi-actuated Ground Vehicles on DWPT Road

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Abstract—The theme proposed for the VTS Motor Vehicle Challenge 2027 (MVC2027) is to promote the understanding of future e-mobility systems and to emphasize the necessity of addressing their energy-related issues through a transdisciplinary approach. To this end, MVC2027 provides participants with a simulator that accurately captures the dynamics of an in-wheel-motor vehicle developed and used at the University of Tokyo. The vehicle is assumed to operate autonomously on roads installed with dynamic wireless power transfer (DWPT) infrastructure. Under realistic conditions, the vehicle is subjected to abrupt variations in road friction due to weather changes, as well as unpredictable disturbances such as strong lateral wind. Participants are therefore invited to develop an integrated strategy that coordinates motor torque distribution, d-axis current control, and front steering angle. The proposed strategy is expected to simultaneously achieve multiple objectives, including (i) improving drivetrain efficiency through the minimization of energy consumption; (ii) ensuring safe traction by regulating wheel slip ratios; (iii) enhancing infrastructure utilization by maximizing the energy received from DWPT; and (iv) improving driving comfort by minimizing deviations in lateral acceleration.

Keywords—in-wheel-motor vehicles, energy management, anti-slip, motor loss reduction, wireless power transfer.

I. INTRODUCTION

The transportation sector is undergoing a fundamental transformation driven by the urgent need to reduce carbon emissions and to develop sustainable urban mobility systems. Emerging technologies such as autonomous driving (AD), operation performance management (OPM), and dynamic wireless power transfer (DWPT) are shaping the future of mobility [1]. The development and integration of these technologies require a transdisciplinary approach that combines expertise from electrical, automotive, control, mechanical, and materials engineering, among others.

In response to these societal demands, the e-Mobility research group at the University of Tokyo has developed an experimental vehicle driven by four in-wheel motors (IWMs) (Fig. 1). The group has continuously pursued the advancement of e-Mobility technologies, and recent achievements include: AD through novel traction control [2] and drone-to-vehicle integration [3]; OPM through optimal speed trajectory [4] and torque distribution methods [5]; and DWPT through lateral misalignment estimation and control [6]. We achieved, for the first time in Japan, the implementation of DWPT on a public road [7], and have contributed to the design of a DWPT system for electric buses at the Osaka Expo [8].



Fig. 1. Electric vehicle FPEV-2 Kanon with four in-wheel-motors.

It is now timely to transition these achievements from isolated developments to an integrated system for real-world services. One of our long-term dreams is to realize an autonomous electric vehicle that transports students, staff, and visitors between Kashiwanoha Campus Station and the Kashiwa Campus of the University of Tokyo. The envisioned system involves a multi-actuated vehicle operating on roads equipped with DWPT infrastructure (Fig. 2). Ensuring safe and efficient operation under various uncertainties—such as abrupt changes in road conditions and unpredictable strong lateral winds—is of critical importance.

By sharing the aforementioned challenges, MVC2027 pursues three primary objectives. First, it aims to enhance the understanding of future e-Mobility concepts and their underlying technologies. Second, it highlights the necessity of a transdisciplinary approach to energy management in next-generation mobility systems. Third, it provides participants and the broader community with a versatile simulator that can be utilized for educational purposes in vehicle dynamics, evaluation of research ideas, and promotion of collaborative research. MVC2027 serves as a platform for discussing and advancing the concept of **MESO** (**Management of Energy Efficiency and Safe Operation**) for multi-actuated vehicles.

II. OVERVIEW OF MESO

A. Energy and safety related issues

As shown in Fig. 2, MVC2027 considers an electric vehicle (EV) equipped with 4 IWMs and one electrical power steering (EPS) mechanism. Each IWM is a permanent magnetic synchronous motor (PMSM) and receives the torque command T_i and the d-axis current command $I_{d,i}$ ($i \in [1,4]$). The EPS is given the front steering angle command δ_f . Investigating the energy efficiency and safe operation of the vehicle, it is essential to note the following:

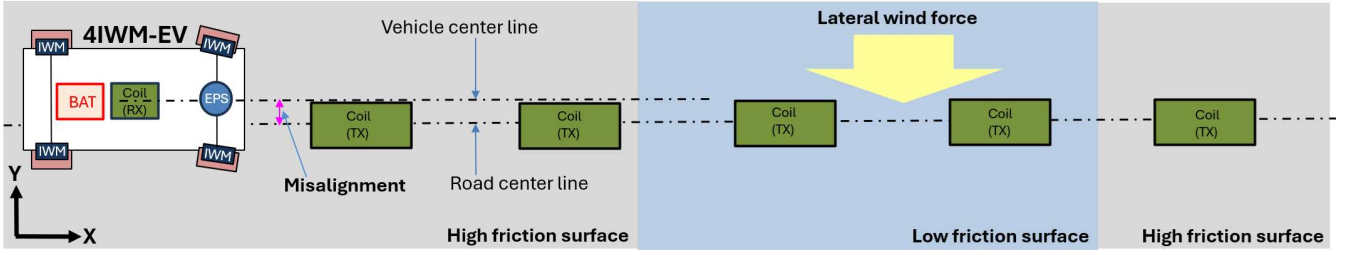


Fig. 2. Multi-actuated vehicle operates on DWPT road (X: longitudinal axis, Y: lateral axis).

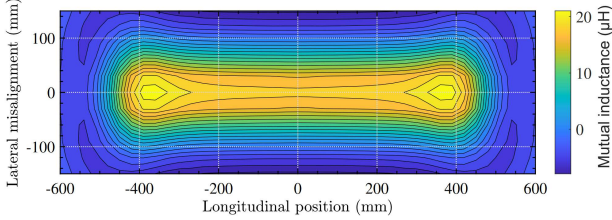


Fig. 3. Mutual inductance map between TX and RX coils.

(1) The power consumption, including motor outputs, copper and iron losses, is a function of T_i and $I_{d,i}$ [5].

(2) The DWPT efficiency strongly depends on the misalignments between the receiver coil (RX) mounted on the vehicle and the transmitter coil (TX) embedded in the road surface (It is assumed that the center line of TX coincides with the center line of the road). As the lateral misalignment increases, the TX-RX mutual inductance L_m decreases, as shown in Fig. 3. If the lateral misalignment exceeds a certain threshold, the DWPT system might be forced to shut down to prevent current overshoot and avoid inefficient operation [9]. It should be noted that lateral misalignment may arise due to strong crosswinds.

(3) Variations in road friction must be considered in energy management. Let V_x and ω_i denote the longitudinal velocity of the vehicle and the i th wheel's rotational velocity, respectively (see Fig. 4). The slip ratio is defined as:

$$\lambda_i = \frac{r\omega_i - V_x}{\max\{r\omega_i, V_x, \varepsilon\}} \quad (1)$$

where r is the wheel radius and ε is a small positive number that prevents division by zero. When operating on low-friction surface, both the slip ratio and wheel velocity might increase considerably, resulting in a deterioration of vehicle traction safety, leading to higher energy consumption, since the motor output power is given by $P_{m,i} = T_i\omega_i$. In addition, the tire lateral force is also limited by road friction (see Fig. 5).

B. Idea of MESO

Following the above discussion, the control of longitudinal and lateral motion is essential not only for safe operation but also for energy efficiency, particularly in the context of DWPT. However, existing studies have largely addressed the aforementioned issues (1)~(3) separately. For instance, [5] considers only torque distribution under straight-line driving. Although lateral misalignment is regulated using EPS in [6], this works does not account for the longitudinal dynamics. Moreover, lateral misalignment control alone is insufficient to maintain desired trajectory. It is also necessary to regulate the yaw angle and yaw-rate by utilizing not only EPS but also the yaw moment generated by the driving force difference between the left and right wheels. While [2] presents an

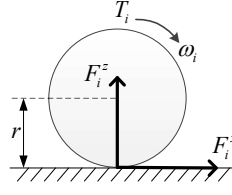


Fig. 4. Wheel rotational motion.

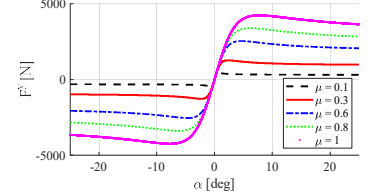


Fig. 5. Lateral tire force characteristics.

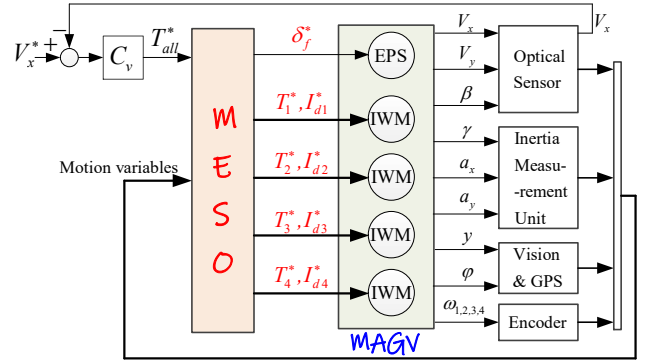


Fig. 6. MESO system for MAGV.

advanced driving force control, it is developed independently of lateral motion and optimal torque/current distribution.

The aforementioned gaps motivate the development of MESO for multi-actuated ground vehicle (MAGV), as generally described in Fig. 6. Given a reference velocity V_x^* , the velocity controller C_v generates the total driving command T_{all}^* . It is assumed that appropriate measurements of the key motion variables are available. Specifically, optical sensors provide measurement of V_x , lateral velocity V_y , and sideslip angle β . An inertia measurement unit provides the yaw-rate γ , as well as longitudinal and lateral acceleration a_x and a_y . The fusion of vision system and global positioning system (GPS) can enable the understanding of lateral misalignment y and yaw angle ϕ . In addition, ω_i is measured by the encoder.

Given T_{all}^* and the measured motion variables, MESO generates the reference values $\{\delta_f^*, T_i^*, I_{d,i}^*\}$ for the front EPS and IWM motor drives. Note that T_i^* is used to derive the corresponding q-axis current of each IWM. The objective of MESO is to improve the efficiency of both drivetrain and DWPT while ensuring safe traction by maintaining appropriate slip ratios. In addition, driving comfort is preserved by constraining the vehicle's lateral acceleration.

III. VEHICLE AND DWPT MODEL

A. Vehicle dynamics model

Let m , I_z , d , l_f and l_r denote the vehicle mass, yaw moment of inertia, track width, and the distances from center of gravity (CG) to the front and rear axles, respectively. As shown in Fig. 7, the lateral dynamics of the vehicle body is:

$$\begin{cases} mV_x(\dot{\beta} + \gamma) = \sum_{1,2} (F_i^x \sin \delta_f + F_i^y \cos \delta_f) + \sum_{3,4} F_i^y + F_d^y \\ I_z \dot{\gamma} = \sum_{1,2} l_f (F_i^x \sin \delta_f + F_i^y \cos \delta_f) - \sum_{3,4} l_r F_i^y + N^z + N_d^z \\ \dot{y} = (V_x / \cos \beta) \sin(\beta + \varphi) \approx V_x(\beta + \varphi); \quad \dot{\varphi} = \gamma \end{cases} \quad (2)$$

where F_d^y and N_d^z are the disturbance force and disturbance yaw moment, which include the effect of lateral wind. The yaw moment N^z is:

$$N^z = \frac{d}{2} (F_2^x - F_1^x) \cos \delta_f + \frac{d}{2} (F_4^x - F_3^x) \quad (3)$$

The longitudinal dynamics is given as:

$$\begin{aligned} m\dot{V}_x &= \sum_{1,2} (F_i^x \cos \delta_f - F_i^y \sin \delta_f) + \sum_{3,4} F_i^x \\ &\quad - \frac{1}{2} \sigma C_D A_F V_x^2 + V_x \gamma \end{aligned} \quad (4)$$

where σ is the air density, C_D is the aerodynamical drag coefficient, and A_F is the frontal area. Let J_w be the wheel inertia and assume that tire rolling resistance is small, the motion of the wheel in Fig. 4 is expressed as follows:

$$J_w \dot{\omega}_i = T_i - r F_i^x \quad (5)$$

The front and rear tire slip angles are expressed as:

$$\alpha_{1(2)} = \tan^{-1} \left(\frac{\beta V_x + \gamma l_f}{V_x \mp \gamma d / 2} \right) - \delta_f; \quad \alpha_{3(4)} = \tan^{-1} \left(\frac{\beta V_x - \gamma l_f}{V_x \mp \gamma d / 2} \right) \quad (6)$$

There exists nonlinear relationships between the tire forces F_i^x , F_i^y and λ_i , α_i [10], which are hidden from participants.

B. IWM power consumption model

The power model of the PMSM is described in steady-state using the set of nomenclatures in Table I. Based on the d-q equivalent circuit in Fig. 8, we have [11]:

$$I_{cd,i} = -\frac{\omega_{e,i} \rho_i L_{d,i} I_{oq,i}}{R_{c,i}}, \quad I_{cq,i} = \frac{\omega_{e,i} (\Psi_{a,i} + L_{d,i} I_{od,i})}{R_{c,i}} \quad (7)$$

$$I_{od,i} = I_{d,i} - I_{cd,i}, \quad I_{oq,i} = I_{q,i} - I_{cq,i} \quad (8)$$

$$T_i = p_{n,i} \Psi_{eq,i} I_{oq,i} \quad (9)$$

$$P_{m,i} = \omega_i T_i; \quad P_{cu,i} = R_{a,i} (I_{d,i}^2 + I_{q,i}^2); \quad P_{fe,i} = R_{c,i} (I_{cd,i}^2 + I_{cq,i}^2) \quad (10)$$

The inverter loss and other losses are assumed to be uncontrollable. Thus, the power consumed by all motors can be considered as $P_{all} = \sum_{i=1}^4 (P_{m,i} + P_{cu,i} + P_{fe,i})$.

C. DWPT efficiency

The S-S equivalent circuit model in Fig. 9 was considered, where the inductance and resistance of the TX and RX coils are L_1, R_1 and L_2, R_2 . The load resistance is denoted by R_L . The resonant capacitors are C_1 and C_2 . The DWPT efficiency is derived as follows [9]:

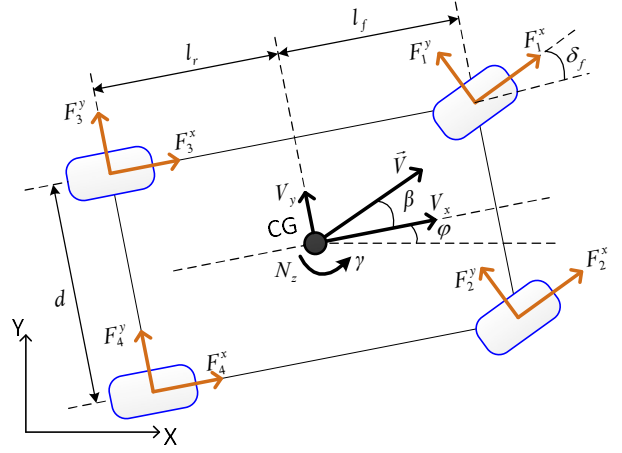


Fig. 7. Planar vehicle model.

TABLE I. NOMENCLATURES TO DESCRIBE MOTOR POWER MODEL

$P_{m,i}$	Output power of motor i
$P_{cu,i}, P_{fe,i}$	Copper loss and iron loss of motor i
$I_{d,i}, I_{q,i}$	The d- and q-axis armature currents of motor i
$I_{cd,i}, I_{cq,i}$	The d- and q-axis iron loss currents of motor i
$I_{od,i}, I_{oq,i}$	$I_{od,i} \triangleq I_{d,i} - I_{cd,i}, I_{oq,i} \triangleq I_{q,i} - I_{cq,i}$
$U_{d,i}, U_{q,i}$	The d- and q-axis terminal voltages of motor i
$L_{d,i}, L_{q,i}$	The d- and q-axis armature self-inductance
ρ_i	Salient coefficient of motor i : $\rho_i \triangleq L_{q,i} / L_{d,i}$
$R_{a,i}$	Armature winding resistance of motor i
$R_{c,i}$	Iron loss resistance of motor i
$\Psi_{e,i}$	Flux linkage of permanent magnet per-phase of motor i
$\Psi_{a,i}$	$\Psi_{a,i} \triangleq \sqrt{3} \Psi_{e,i}$
$\Psi_{eq,i}$	$\Psi_{eq,i} \triangleq \Psi_{a,i} + (1 - \rho_i) L_{d,i} I_{od,i}$
$p_{n,i}$	Number of pole pairs of motor i
$\omega_{e,i}$	Electrical angular velocity of motor i ($\omega_{e,i} \triangleq p_{n,i} \omega_i$)

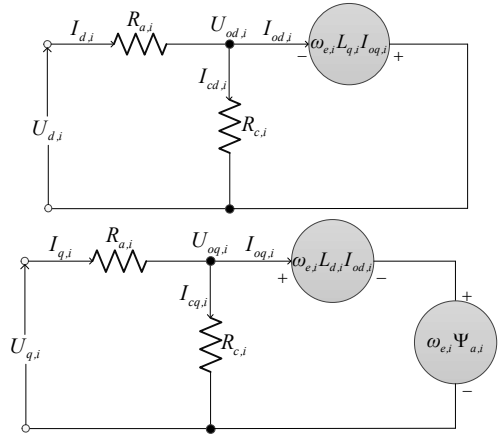


Fig. 8. The d-q equivalent circuit of the PMSM.

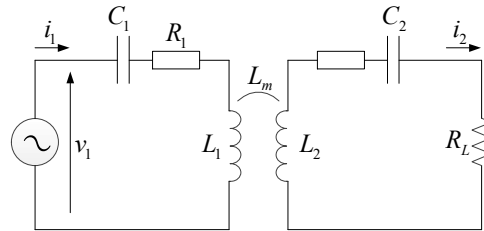


Fig. 9. S-S equivalent circuit model for DWPT.

$$\eta_{DWPT} = \frac{(\omega_0 L_m)^2}{R_1 R_2 + R_1 R_L + (\omega_0 L_m)^2} \frac{R_L}{R_2 + R_L} \quad (11)$$

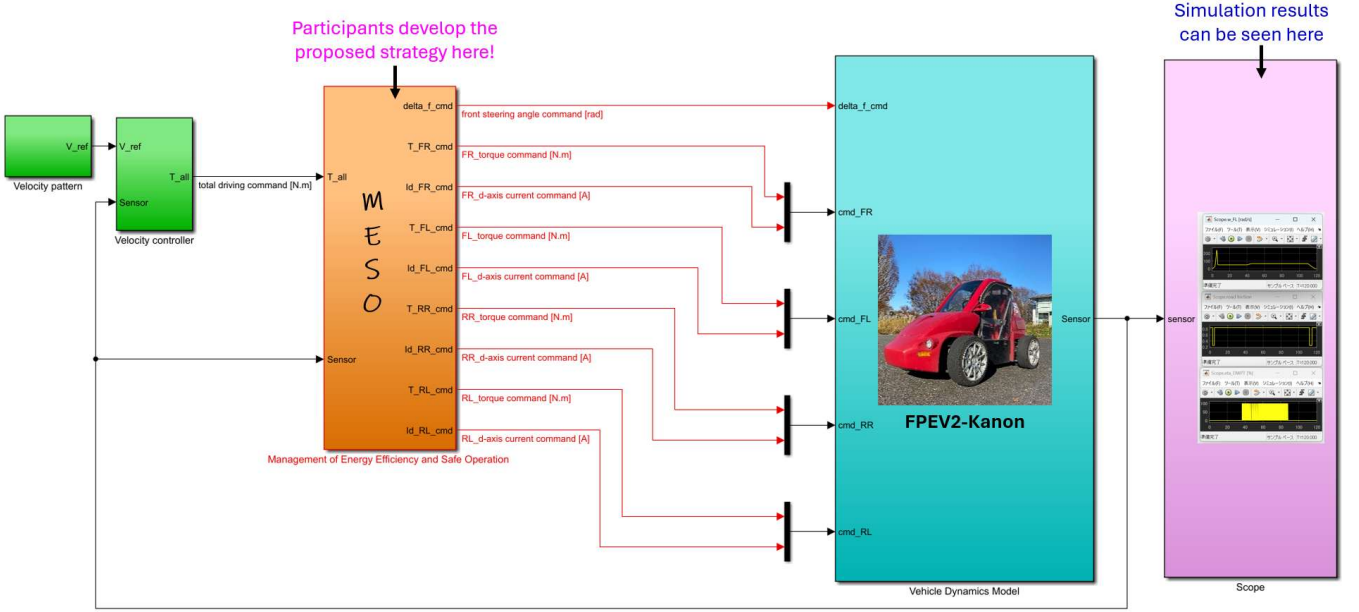


Fig. 10. Simulink model of MAGV simulator.

IV. DESCRIPTION OF THE MVC2027

A. General problem setting

(1) Simulator: Based on the MAGV in Fig. 1, a simulator was developed using MATLAB R2022b (Fig. 10). The main parameters are summarized in Table II. The sampling time of the simulator is set to 0.2 ms. A velocity controller was provided, and participants are invited to implement the MESO block.

(2) Testing scenario: MVC2027 considers straight-road driving only, without cornering. As shown in Fig. 11, the reference velocity profile consists of several segments, including acceleration, deceleration, and constant velocity phases. Lateral wind force attacks the vehicle from 45 to 55 seconds. The external disturbances F_d^y , N_z^d , and the road friction coefficient μ are assumed to be time variable and unknown. Therefore, the participants must estimate these variables if they intend to utilize them in the MESO algorithm. It should be note that the proposed algorithm will be evaluated not only by the driving cycle in Fig. 11, but also by an additional “secret” driving cycle.

(3) DWPT: With respect to the speed profile in Fig. 11(a), the driving distance is approximately 2020 m. The resonant frequency is set to 85 kHz. The TX coils are embedded along the center line of the road surface from 500 m to 1500 m. Each TX coil has a length of 1000 mm, and the spacing between consecutive coils is also 1000 mm. The transmitted power by TX is regulated at $P_{TXn} = 10\text{kW}$.

Remark-1: In the problem setting of MVC2027, roll and pitch motions, suspension mechanism, battery model, as well as inverter switching behavior, are not considered.

B. Scoring

As explained in Section II, MESO aims to attain multiple objectives, which are described as follows:

(1) Efficiency of drivetrain, from the beginning until the finish time t_{end} , is achieved by minimizing the cost function:

$$\Phi_1 = \int_0^{t_{end}} P_{all} dt = \int_0^{t_{end}} \sum_{i=1}^4 (P_{m,i} + P_{cu,i} + P_{fe,i}) dt \quad (12)$$

(2) Safe traction is achieved by minimizing the slip ratios:

$$\Phi_2 = \int_0^{t_{end}} \sum_{i=1}^4 \lambda_i^2 dt \quad (13)$$

(3) Efficiency of DWPT is achieved by minimizing:

$$\Phi_3 = \int_0^{t_{end}} (1 - \eta_{DWPT})^2 dt \quad (14)$$

(4) Driving comfort is achieved by to minimizing the deviation of lateral acceleration:

$$\Phi_4 = \int_0^{t_{end}} a_y^2 dt \quad (15)$$

In summary, the following scoring function is defined to evaluate the performance of the proposed strategies:

$$\Phi = \sigma_1 \Phi_1 + \sigma_2 \Phi_2 + \sigma_3 \Phi_3 + \sigma_4 \Phi_4 \quad (16)$$

where σ_i is the weight assigned to each term in (16).

C. Constraints

In addition, the following constraints are considered:

(1) Total driving command: $\sum_{i=1}^4 T_i^* = T_{all}^*$, where is given by the velocity controller.

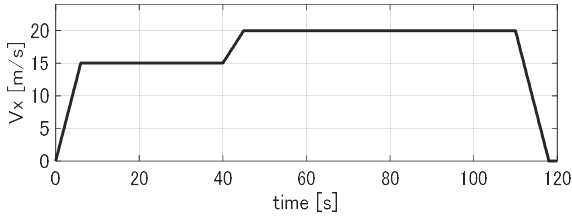
(2) Torque limitation: $|T_i^*| \leq T_{max,i}, \forall i \in [1,4]$.

(3) Friction limit circle: $\sqrt{(F_i^x)^2 + (F_i^y)^2} \leq \mu F_i^z, \forall i \in [1,4]$ where F_i^z is the vertical force acting at each wheel.

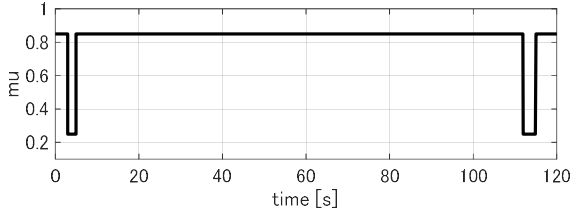
(4) Coil misalignment: Wireless power transfer is enabled only when $|\Delta_x| \leq 380 \text{ mm}$ and $|\Delta_y| \leq 70 \text{ mm}$, where Δ_x and Δ_y denote the longitudinal and lateral misalignments between TX and RX coils, respectively. This is to ensure sufficient mutual inductance during power transfer operation.

TABLE II. SPECIFICATION OF ELECTRIC VEHICLE KANON

Vehicle mass & wheel radius	$m = 880 \text{ kg}; r = 0.302 \text{ m}$
Inertia of the wheel	$J_{w,1(2)} = 1.24 \text{ kg} \cdot \text{m}^2$ $J_{w,3(4)} = 1.26 \text{ kg} \cdot \text{m}^2$
Distance from CG to wheel axes	$l_{f/r} = 0.988/0.712 \text{ m}$
Frontal area of the vehicle	$A_F = 2.40 \text{ m}^2$
Front motor	$T_{max,1(2)} = 500 \text{ N} \cdot \text{m}$ $R_{a,1(2)} = 0.086 \Omega$ $L_{d,1(2)} = 0.0045 \text{ H}$ $L_{q,1(2)} = 0.0049 \text{ H}$ $\Psi_{a,1(2)} = 0.180 \text{ Wb}$ $p_{n,1(2)} = 10$
Rear motor	$T_{max,3(4)} = 530 \text{ N} \cdot \text{m}$ $R_{a,3(4)} = 0.143 \Omega$ $L_{d,3(4)} = 0.00130 \text{ H}$ $L_{q,3(4)} = 0.00150 \text{ H}$ $\Psi_{a,3(4)} = 0.125 \text{ Wb}$ $p_{n,3(4)} = 12$
DWPT	$L_1 = 142.32 \times 10^{-6} \text{ H}$ $L_2 = 54.20 \times 10^{-6} \text{ H}$ $R_1 = 85.02 \times 10^{-3} \Omega$ $R_2 = 39.88 \times 10^{-3} \Omega$ $R_k = 5.7 \Omega$



(a) Speed pattern



(b) Variation of road friction coefficient

Fig. 11. Testing scenario.

V. SIMULATION DEMONSTRATION

A. Examples of MESO strategies

Two examples of MESO strategies are provided to participants:

(1) **Baseline strategy (BS)**: The torque are equally distributed to all IWMs, which mean the vehicle is not provided with yaw moment control. The d-axis current is $I_{d,i}^* = 0$ for all IWMs. The front EPS angle is generated by the following control law:

$$\delta_f^* = K_{\delta y,p} e_y + K_{\delta \varphi,p} e_\varphi \quad (17)$$

where $e_y = y^* - y$, $e_\varphi = \varphi^* - \varphi$ with the reference values $y^* = 0$ and $\varphi^* = 0$. The control gains are $K_{\delta y,p}$ and $K_{\delta \varphi,p}$.

(2) **Improved strategy (IS)**: The front EPS angle is generated similarly to (17). Besides, the IWM torques are distributed based on a yaw moment control law as follows:

$$T_{1,3}^* = \frac{T_{all}^*}{4} - \frac{r}{2d} N^{z*}; T_{2,4}^* = \frac{T_{all}^*}{4} + \frac{r}{2d} N^{z*} \quad (18)$$

$$N^{z*} = K_{N_y,p} e_y + K_{N_y,i} \int_0^t e_y dt + K_{N_y,d} \frac{de_y}{dt} + K_{N_\varphi,p} e_\varphi \quad (19)$$

with the control gains $K_{N_y,p}$, $K_{N_y,i}$, $K_{N_y,d}$, and $K_{N_\varphi,p}$. Assuming that the motor salient coefficients are close to 1, the d-axis current that minimizes motor losses is given by the following formula [11]:

$$I_{od,i}^* = -\frac{\omega_{e,i}^2 L_{d,i} (R_{a,i} + R_{c,i}) \Psi_{a,i}}{R_{a,i} R_{c,i}^2 + \omega_{e,i}^2 L_{d,i}^2 (R_{a,i} + R_{c,i})} \quad (20)$$

B. Results and discussions

The simulation results of the BS are presented in Fig. 12. The BS fails to prevent wheel slip when the vehicle accelerates on a low-friction surface. In addition, under strong crosswind disturbances, the vehicle exhibits significant lateral misalignment, leading to temporary deactivation of the DWPT function. The total energy consumption under the BS is 323.5 Wh.

The simulation results of the IS are shown in Fig. 13. By differentiating the left and right motor torques, it generates an additional yaw moment that suppresses lateral misalignment. Furthermore, the energy consumption is reduced to 319.2 Wh by controlling the negative d-axis currents of the IWMs. However, the IS remains a relatively simple strategy and is still unable to prevent slip under low-friction conditions. Although the vehicle can harvest more energy from the DWPT, the lateral acceleration deviation increases, indicating a potential trade-off between DWPT efficiency and driving comfort.

VI. CONCLUSIONS

In this paper, the MESO framework for the IEEE VTS MVC2027 has been presented, with a focus on the motion control and electrification of multi-actuated vehicles. Participants are invited to develop integrated strategies that address the trade-offs among multiple objectives. All relevant materials will be available in the MVC2027 GitHub repository. The provided simulator serves as a realistic testbed that supports both research and educational activities, contributing to the advancement of future e-Mobility systems. Furthermore, the simulator is designed to be extensible, allowing users to incorporate additional models, ranging from vehicle body dynamics to battery and inverter systems.

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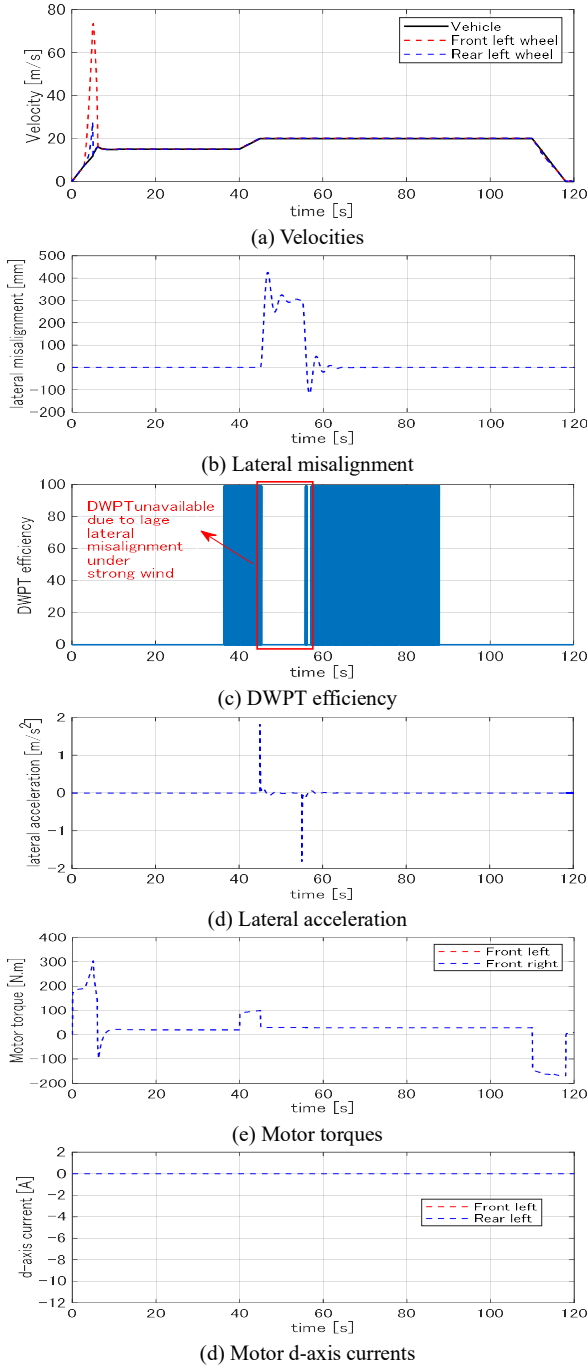


Fig. 12. Baseline strategy simulation results.

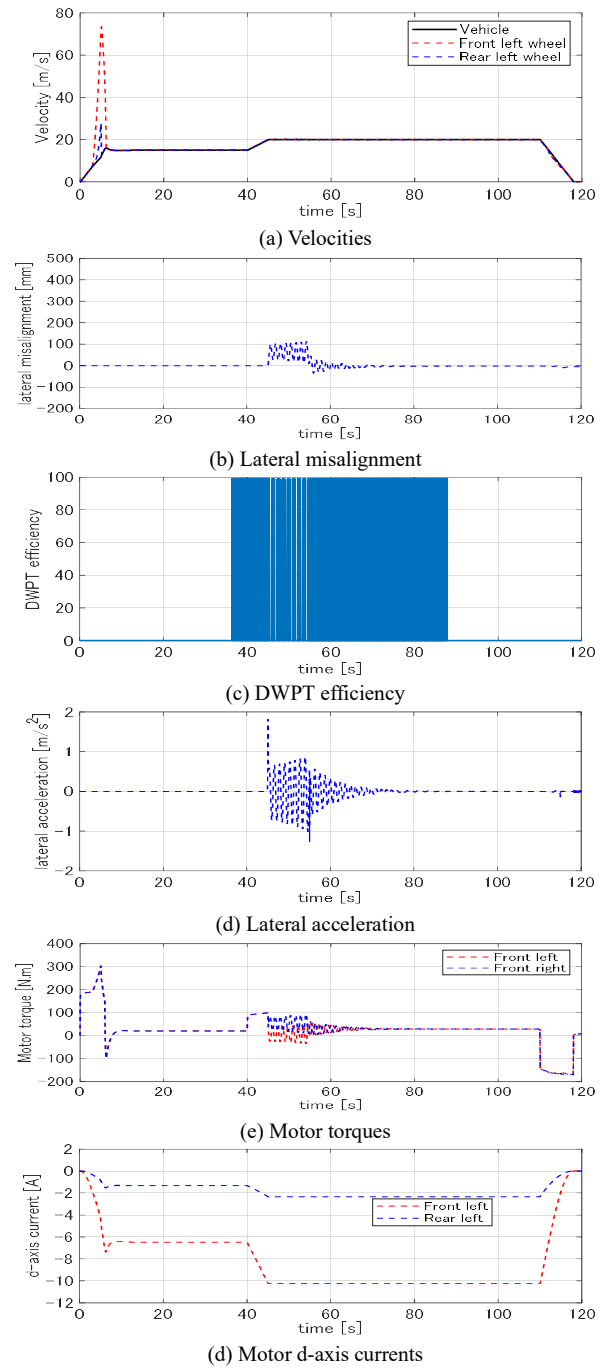


Fig. 13. Improved strategy simulation results.

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